

definitions

 **image** *a static snapshot of container's configuration.*

 **container** *an application sandbox. each container is based on an image.*

 **layer** *image is composed of read-only file system layers. container creates single writable layer.*

 **docker registry** *remote server for storing Docker images*

 **Dockerfile** *a configuration file with build instructions for Docker images*

 **docker engine** *Docker platform installation running on a given host*

 **docker client** *client application that talks to local or remote Docker daemon*

 **docker daemon** *service process that listens to Docker client commands over local or remote network*

 **docker host** *server that runs Docker engine*

 **volume** *directory shared between host and container*

docker run

docker run [OPTIONS] IMAGE[:TAG] [COMMAND]

Run a command in a new container.

metadata

--name=CNTR_NAME Assign a name to the container.
-l, --label NAME[=VALUE] Set metadata on the container.

process

-d, --detach Run in the background.
-i, --interactive Keep STDIN open.
-t, --tty Allocate a pseudo-TTY.
--rm Automatically remove the container when process exits.
-u USER Run as username or UID.
--privileged Give extended privileges.
-w DIR Set working directory.
-e NAME=VALUE Set environment variable.
--restart=POLICY Restart policy.
 no on-failure[:RETRIES]
 always unless-stopped

network

-P, --publish-all Publish all exposed ports to random ports.
-p HOST_PORT:CNTR_PORT Expose a port or a range of ports.
--network=NETWORK_NAME Connect container to a network.
--dns=DNS_SERVER1[,DNS_SERVER2] Set custom dns servers.
--add-host=HOSTNAME:IP Add a line to /etc/hosts.

file system

--read-only Mount the container's root file system as read only.
-v, --volume [HOST_SRC:]CNTR_DEST Mount a volume between host and the container file system.
--volumes-from=CNTR_ID Mount all volumes from another container.

Dockerfile

FROM <image_id>
 base image to build this image from

RUN <command> *shell form*
RUN ["<executable>",
 "<param1>", *exec form*
 ...,
 "<paramN>"]
 executes command to modify container's file system state

MAINTAINER <name>
 provides information about image creator

LABEL <key>=<value>
 adds searchable metadata to image

ARG <name>[=<default value>]
 defines overridable build-time parameter:
 docker build --build-arg <name>=<value> .

ENV <key>=<value>
 defines environment variable that will be visible during image build-time and container run-time

ADD <src> <dest>
 copies files from <src> (file, directory or URL) and adds them to container file system under <dest> path

COPY <src> <dest> *similar to ADD, does not support URLs*

VOLUME <dest>
 defines mount point to be shared with host or other containers

EXPOSE <port>
 informs Docker engine that container listens to port at run-time

WORKDIR <dest>
 sets build-time and run-time working directory

USER <user>
 defines run-time user to start container process

STOPSIGNAL <signal>
 defines signal to use to notify container process to stop

ENTRYPOINT *shell form* or *exec form*
 defines run-time command prefix that will be added to all run commands executed by docker run

CMD *shell form* or *exec form*
 defines run-time command to be executed by default when docker run command is executed